Autonomous Driving Technology, Cloud Computing, and Scalability Benchmark

Tesla, Waymo, NVIDIA, and Mobileye

Executive Summary

Historical cutoff: June 30, 2025
Independent public-data reconstruction
Jannatul Nayeem | July 2026

Decision Context

This portfolio project reconstructs and extends an earlier Microsoft client based global internship project that was facilitated through the Russell Palmer Career Management Center at Michigan State University. It is an independent public data reconstruction. Jannatul Nayeem was not employed by Microsoft, and these files are not represented as the exact historical client deliverables.

The benchmark is role aware. Tesla, Waymo, NVIDIA, and Mobileye are compared only where the underlying concept, unit, period, automation level, and exposure are reasonably aligned.

Ecosystem Roles

Company	Primary role	Scale pathway	Comparison boundary
Tesla	Vehicle manufacturer and consumer supervised ADAS developer	Direct consumer fleet distribution; vertically integrated vehicle, software, and inference hardware	Supervised Level 2 exposure is not comparable to rider only Level 4 service mileage
Waymo	Level 4 ADS developer and mobility operator	Geofenced rider only service; operational fleet and matched safety research	Deployment is limited to defined operating domains and selected cities
NVIDIA	Compute, software, simulation, and reference platform provider	Train, simulate, and deploy stack for OEM and mobility partners	Partner fleet incidents or miles cannot be attributed directly to NVIDIA
Mobileye	OEM ADAS and autonomy technology supplier	Purpose built EyeQ platform, mapping, and product tiers from base ADAS to higher autonomy	Systems shipped span generations, customers, and automation levels

Source control: Role classifications are grounded in Tesla 2024 Form 10-K, Waymo dated research and deployment releases, NVIDIA FY2025 Form 10-K, and Mobileye 2024 Form 10-K. See SRC009, SRC010, SRC015, SRC018, and SRC023.

Most Supported Findings

- **Waymo safety evidence.** The May 2025 matched benchmark study examined 56.7 million rider only miles through January 2025. It reported statistically significant reductions for selected safety relevant outcomes. The result applies to Waymo operating domains and the study methodology; it does not establish a universal industry ranking.
- **Tesla distribution and learning channel.** Tesla had the broadest direct consumer vehicle distribution channel in this group. Its public filings describe vision based self driving development, custom inference hardware, over the air software, and internal AI infrastructure. Public exposure by software version and operating domain was not available in a form suitable for cross company normalization.
- **NVIDIA platform breadth.** NVIDIA disclosed an end to end train, simulate, and deploy platform that combines accelerated data center computing, Omniverse based simulation, DRIVE AGX, DriveOS, and reference sensor architectures. Its scalability is primarily indirect through OEM and mobility partners.
- **Mobileye OEM scale.** Mobileye reported approximately 29.0 million EyeQ and SuperVision systems shipped in 2024 and SoCs deployed in more than 200 million vehicles by year end. These figures demonstrate integration scale but should not be interpreted as autonomous vehicle deployment.
- **Safety comparability.** Raw NHTSA or California totals are insufficient for direct safety comparison because reporting thresholds, telemetry, exposure, geography, vehicle type, software version, and human intervention policies differ.

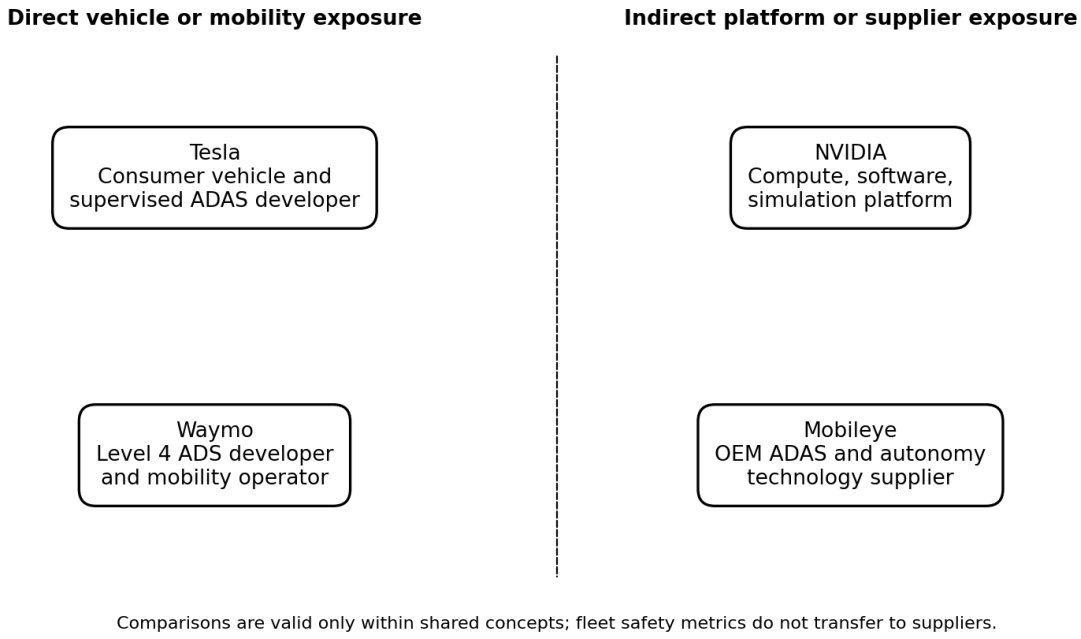


Figure 1. Role aware comparison framework.

Source control: Selected quantitative evidence: SRC009, SRC010, SRC015, SRC018, SRC019, and SRC023. Regulatory comparability rules: SRC001 through SRC004, SRC034, and SRC035.

Safety and Deployment Evidence

Indicator	Public value	Interpretation	Source
Waymo rider only miles analyzed	56.7 million miles	Through January 2025; matched human benchmarks	SRC009
Waymo paid trips	More than 250,000 per week	Point in time, May 2025	SRC010
Waymo operating vehicles	More than 1,500 vehicles	Point in time, May 2025	SRC010
Tesla vehicle deliveries	1,789,226 in 2024	Vehicle deliveries, not autonomous exposure	SRC015
Mobileye systems shipped	Approximately 29.0 million in 2024	Primarily EyeQ SoCs; mixed ADAS tiers	SRC023
California AV test miles	4,498,066 in the 2024 reporting period	Program total across permit holders	SRC006

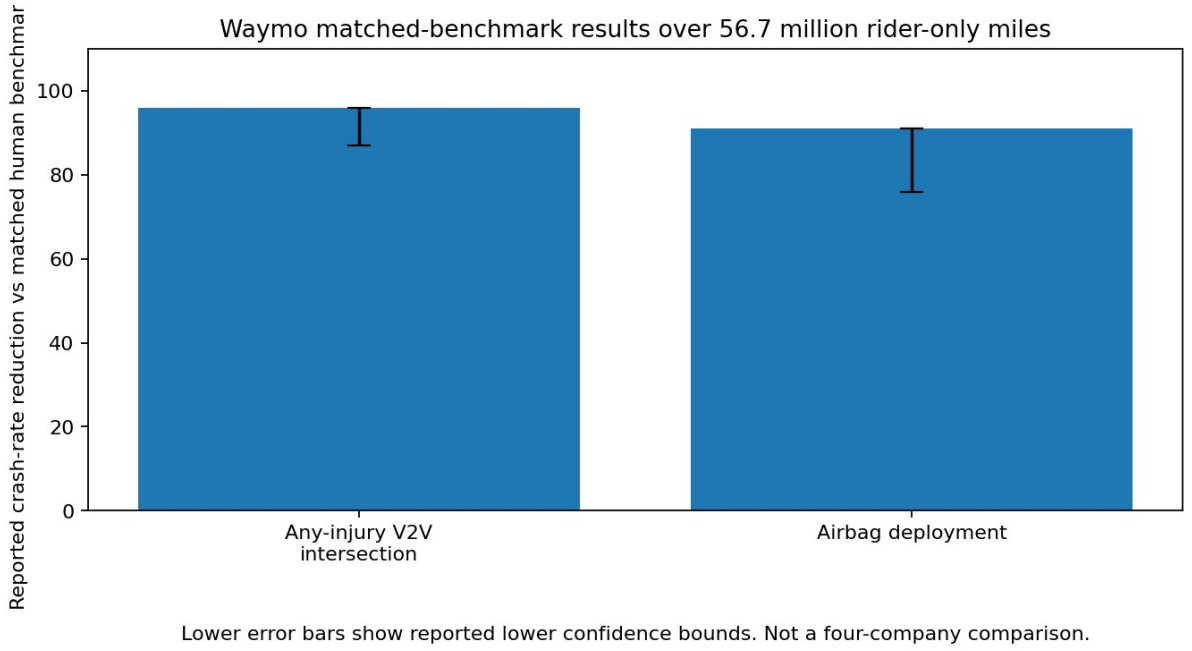


Figure 2. Selected matched Waymo benchmark results as reported in the May 2025 study.

Why Incident Totals Cannot Be Ranked Directly

- ADS and Level 2 reporting thresholds are not identical.
- NHTSA records known reportable incidents, not complete exposure denominators.
- Companies differ in telemetry coverage and their ability to learn that an incident occurred.
- A geofenced rider only Level 4 fleet is not the same risk population as nationwide supervised consumer driving.
- Software, hardware, road type, weather, time, and geography are often unavailable or not aligned.

Normalization is defensible only when the event definition, exposure mileage, operating domain, period, geography, and severity measure are aligned. Otherwise, the output remains descriptive.

Cloud and GPU Cost Scenarios

The cloud model standardizes workloads across AWS, Microsoft Azure, and Google Cloud historical pricing references. It is not an estimate of confidential company spending. Each scenario includes training compute, simulation compute, persistent storage, checkpoint storage, egress, utilization, and an explicit price factor.

Scenario	Lowest on demand	Lowest committed assumption	Lowest interruptible assumption
Small experimentation	$1,227	$881	$593
Medium development	$109,706	$77,447	$50,564
Large AV training	$2,858,251	$2,012,476	$1,307,663

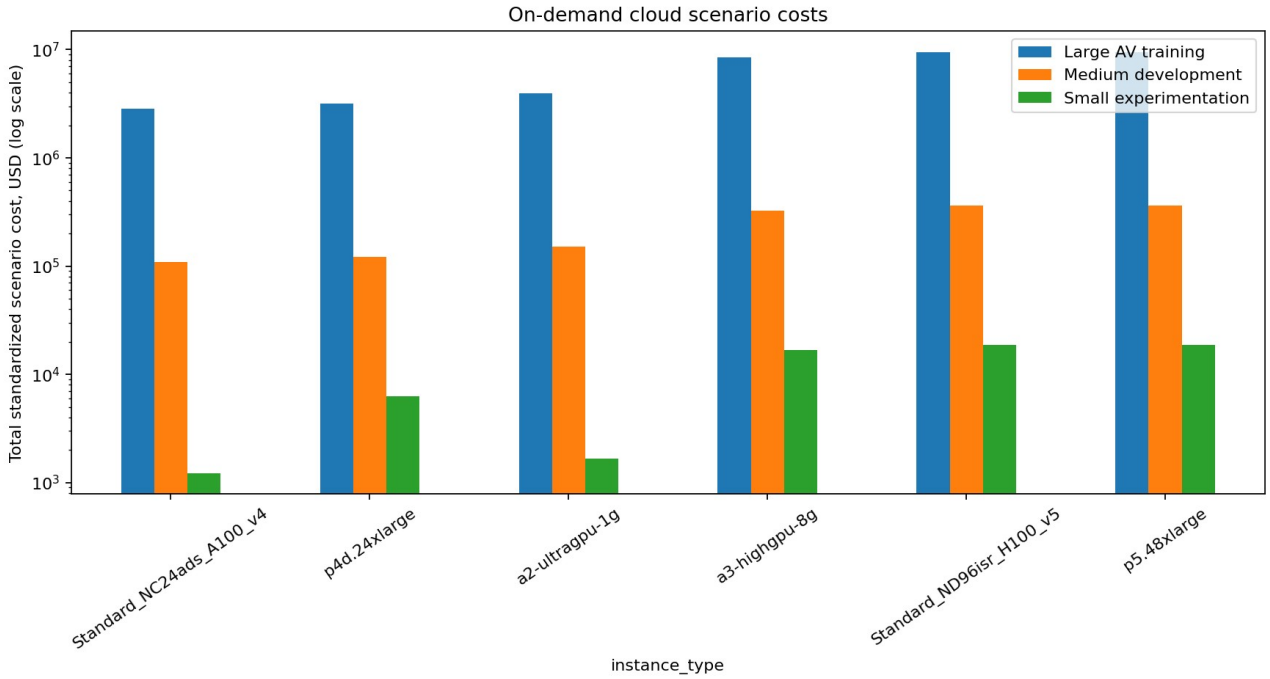


Figure 3. Standardized scenario cost by provider and pricing treatment.

Primary Cost Drivers

- Required GPU hours and the number of repeated experiments.
- Utilization, because inefficient provisioning increases paid wall clock time.
- Instance price and the assumed on demand or discounted treatment.
- Simulation workload, particularly in the large autonomous driving scenario.
- Storage, checkpoint retention, and outbound transfer at high data volumes.

Source control: GPU inputs are logged under SRC026 through SRC030. Discount factors of 1.00, 0.70, and 0.45 are explicit scenario assumptions, not quoted provider discounts.

Scalability Framework and Portfolio Value

The model uses twelve criteria scored on a 1 to 5 ordinal scale with three alternative weighting schemes. The scores make evidence and tradeoffs visible; they do not create a universal winner because the companies occupy different roles.

Criterion	Definition	Baseline weight	Measurement approach
Data scale	Breadth, frequency, and relevance of accessible driving data	10	Evidence of deployed/test data channels; not raw vehicle count alone
Training infrastructure	Ability to train and iterate large models	10	Public stack, accelerators, and workflow evidence
Simulation capabilities	Breadth and maturity of simulation/validation	8	Closed-loop and scenario coverage evidence
In-vehicle compute	Capability and efficiency of vehicle compute	8	Production platform and workload fit
Sensor complexity	Manufacturing and operating simplicity	8	Higher score means simpler scalable hardware, not necessarily higher perception redundancy
Geographic scalability	Ability to extend across regions and ODDs	10	Current reach plus adaptation burden
Hardware dependence	Exposure to proprietary or costly hardware constraints	7	Higher score means lower dependency/constraint
OEM or fleet integration	Ability to integrate into vehicles or fleet operations	10	Production integration, reference platforms, fleet processes
Deployment maturity	Evidence of production or commercial deployment	10	Role-specific deployment evidence
Regulatory exposure	Ability to manage regulatory and reporting complexity	7	Higher score means more manageable exposure
Commercial flexibility	Number and flexibility of routes to market	7	Direct, platform, supplier, and partnership options
Cost scalability	Expected cost curve for hardware and operations	5	Relative architecture burden; no confidential unit economics

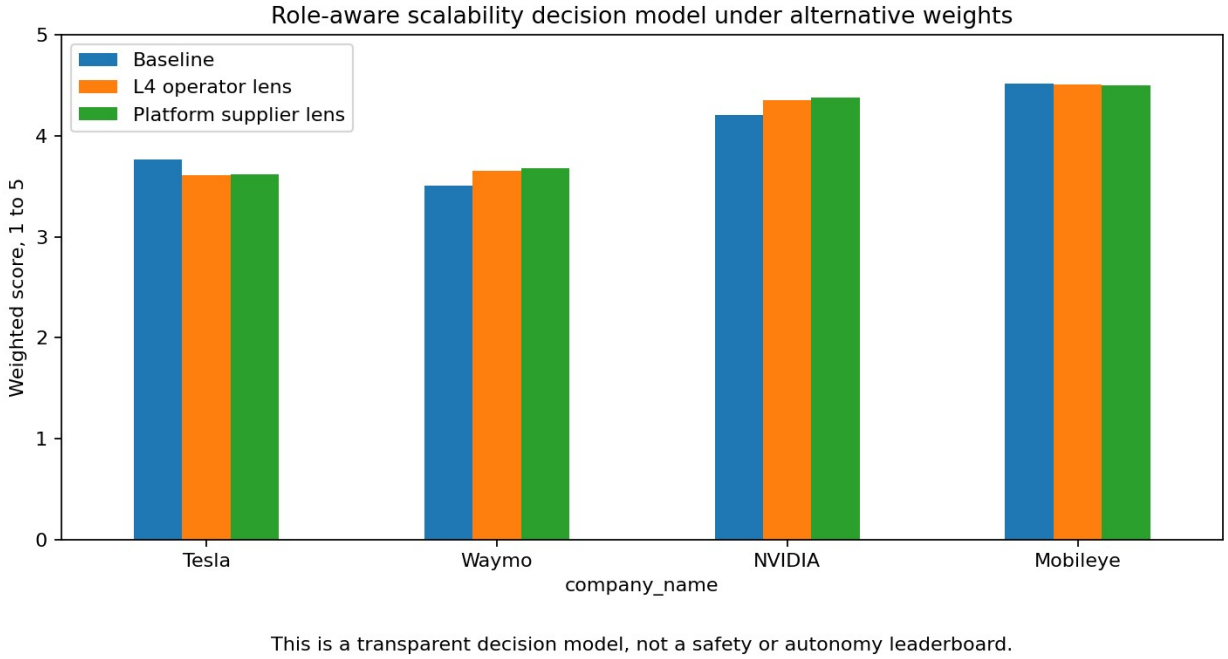


Figure 4. Sensitivity of weighted scores to alternative decision lenses.

A high score can mean a different advantage by role. For example, low sensor complexity can support manufacturing scale, while a more redundant sensor suite can support perception robustness within a constrained Level 4 operating domain.

Delivered Portfolio Assets

- Executed SQLite database with normalized dimensions, facts, bridges, quality results, and reporting views.
- Reproducible Python pipeline and four executed analytical notebooks.
- Thirty five source register entries and twenty documented validation checks.
- Power BI ready workbook, CSV exports, relationship specification, DAX measures, build guide, and four page mockups.
- Written report, methodology, data dictionary, assumptions register, interview guide, and resume outputs.

The strongest conclusion is not that one company wins. The evidence supports four distinct scalability paths: consumer distribution, geofenced mobility operations, horizontal compute and simulation platforms, and OEM technology supply.